



# Understanding the Basic Operations of VRP

Difficulty (out of 5 stars): ★★★

Recommended study time: 2 hours

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# 1 About This Lab

## 1.1 Task Description

Log in to a new device to configure it. A device supports first login through the console port or the mini USB port. Use the console port login mode in this task. After local login, complete basic configuration to provide a foundational environment for subsequent configuration. The basic configuration involves configuring the device name, management IP address, and system time, as well as the Telnet user level and authentication mode to implement remote login. After completing this task, gain a clear understanding of Huawei VRP and be capable of performing basic operations on Huawei devices independently.

## 1.2 Objectives

On completing this task, you will be able to:

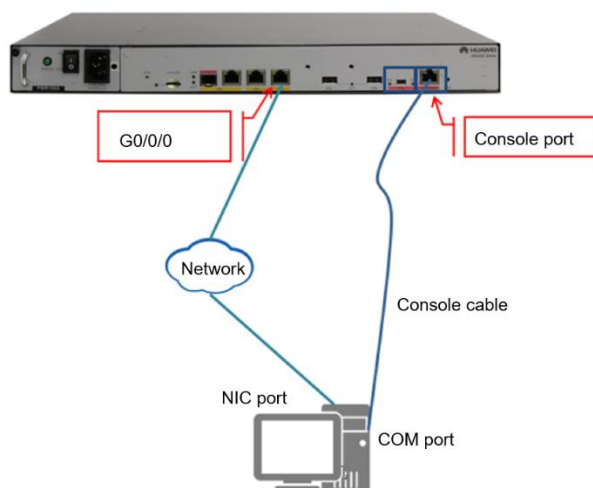
1. Log in to a device for the first time through the console port.
2. Check system information.
3. Modify the system time.
4. Use the help function and command auto-completion function.
5. Change the device name.
6. Perform login configuration.
7. Set console port parameters.
8. Configure interface IP addresses and descriptions.
9. View the file list on a device.
10. Manage the configuration file of a device.
11. Restart a device.
12. Remotely manage a device using Telnet.
13. Remotely manage a device using SSH.



## 2 Task Preparation

### 2.1 Network Topology

Figure 2-1 Lab topology



### 2.2 Initial Configuration

Before logging in to a device through the console port, ensure that you have completed the following tasks:

- Power on the device properly. Prepare a console cable.
- Prepare the PC terminal emulation software. PuTTY is used as an example.

## 3 Task Implementation

### 3.1 Logging In to the Device for the First Time Through the Console Port

Use a console cable to connect the console port of the device with the COM port of a computer. You can then log in to the device and perform local commissioning and maintenance. A console port is an RJ45 port that complies with the RS232 serial port standard. At present, the COM ports provided by most desktop computers can be connected to console ports. In most cases, a laptop does not provide a COM port. Therefore, a USB-to-RS232 conversion port is required if you use a laptop.

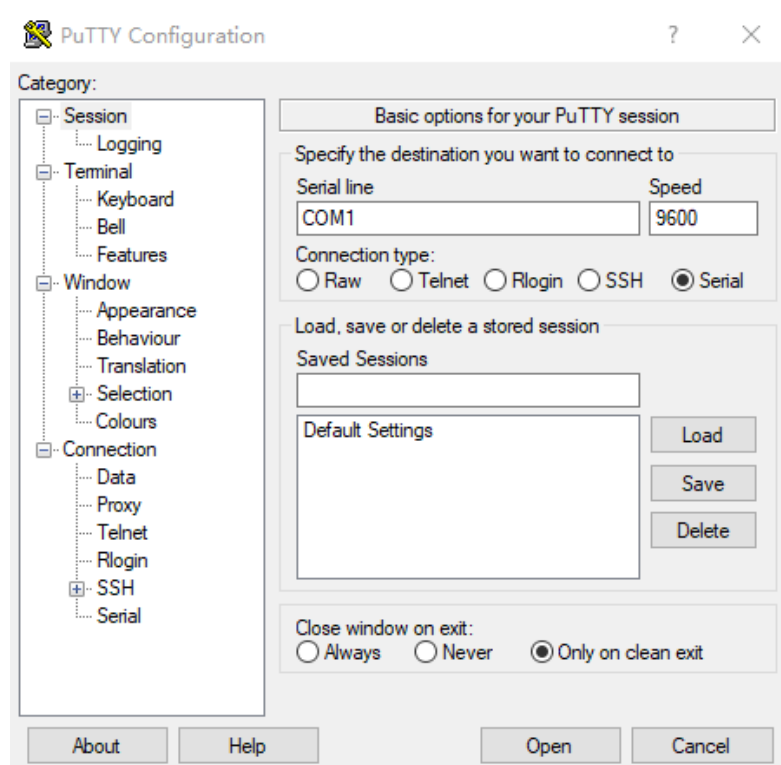
**Step 1** Connect the DB9 connector of the prepared console cable to the PC's COM port, and the RJ45 connector to the device's console port.

```
<R1>system-view
Enter system view, return user view with Ctrl+Z.
[R1]dhcp enable
Info: The operation may take a few seconds. Please wait for a moment.done.
[R1]
```



**Step 2** Start the terminal emulation software on the PC and create a connection. The following uses the third-party software PuTTY as an example, as shown in Figure 3-1.

**Figure 3-1** Creating a connection

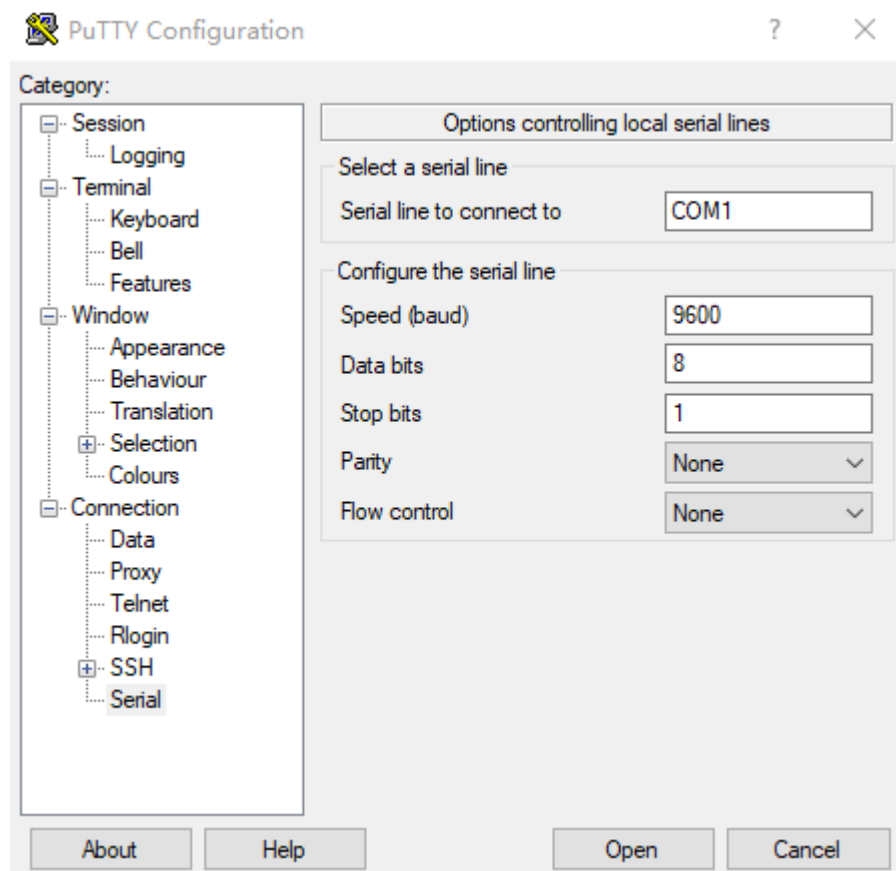


Select the port based on actual situations. For example, on Windows, you can open **Device Manager** to view port information and select the port to be connected. Set the communication parameters. Ensure that the communication parameter settings in the terminal emulation software are consistent with the default parameter settings (9600 bit/s transmission rate, 8 data bits, 1 stop bit, no parity check, and no flow control) of the device's console port.



**Step 3** Set the connection port and communication parameters, as shown in Figure 3-2.

**Figure 3-2** Setting the connection port and communication parameters



**Step 4** Click **Open**. The following information is displayed, asking you to set the login password. The console port allows access without a password by default. Users can directly connect and configure the device.

```
Press any key to get started
Login authentication
Username:admin
Password:
Warning: There is a risk in the current configuration file. Please save configuration as soon as possible.
<Huawei>
Warning: Auto-Config is working. Before configuring the device, stop Auto-Config. If you perform configurations when Auto-Config is running, the DHCP, routing, DNS, and VTY configurations will be lost. Do you want to stop Auto-Config? [y/n]:y
Info: Auto-Config has been stopped.
<Huawei>
```

When you log in to a new device (or a device without startup configuration files) through the console port, the system displays the following message: "Auto-Config is working. Before configuring the device, stop Auto-Config. If you perform configurations when Auto-Config is running, the DHCP, routing, DNS, and VTY configurations will be lost. Do you want to stop



Auto-Config? [y/n]:". If auto configuration is required, enter **n** and press **Enter**. To stop auto configuration, enter **y** and press **Enter**.

If no special requirement is posed, you are advised to select the manual configuration mode.

After logging in to the device through the console port for the first time, you can perform basic configurations, such as configuring the device time and date, system character set encoding, device name, management IP address, and Telnet user level and authentication mode.

----End

## 3.2 Checking the System Information

Run the **display version** command to check the software version and hardware information of a router.

```
<Huawei>display version
Huawei Versatile Routing Platform Software
VRP (R) software, Version 5.130 (AR2200 V200R003C00)
Copyright (C) 2011-2012 HUAWEI TECH CO., LTD
Huawei AR2220 Router uptime is 0 week, 0 day, 0 hour, 0 minute
BKP 0 version information:
1. PCB      Version   : AR01BAK2A VER.NC
2. If Supporting PoE : No
3. Board    Type      : AR2220
4. MPU Slot Quantity : 1
5. LPU Slot Quantity : 6

MPU 0(Master) : uptime is 0 week, 0 day, 0 hour, 0 minute
MPU version information :
1. PCB      Version   : AR01SRU2A VER.A
2. MAB      Version   : 0
3. Board    Type      : AR2220
4. BootROM  Version   : 0

<Huawei>
```

The command output shows the VRP version, device model, and startup time.

## 3.3 Changing the System Time

The VRP system automatically saves the time. If the time is incorrect, you can run the **clock timezone** and **clock datetime** commands in the user view to change the system time.

```
<Huawei>clock timezone Local add 08:00:00
<Huawei>clock datetime 08:49:00 2025-04-03
<Huawei>
```

You can change the **Local** field value to the time zone name of the current region. If the current time zone is located in the west of the UTC+0 time zone, replace the **add** parameter with **minus**.

Run the **display clock** command to check the new system time.

```
<Huawei>display clock
2025-04-03 08:49:10
```



Thursday

Time Zone(Local) : UTC+08:00

&lt;Huawei&gt;

### 3.4 Using the Help Function and Command Auto-Completion Function

When you enter a command in the system, the question mark (?) is a wildcard, and the **Tab** key is the shortcut key for automatically completing commands.

```
<Huawei>display ?
Cellular                Cellular interface
aaa                     AAA
access-user             User access
accounting-scheme       Accounting scheme
acl                     <Group> acl command group
activated-alarm          Activated alarm
actual                  Current actual
alarm                   Alarm
als                     Als
antenna                 Current antenna that outputting radio
anti-attack              Specify anti-attack configurations
ap                      <Group> ap command group
ap-auth-mode             Display AP authentication mode
.....Omitted.....
```

Enter a question mark (?) after entering partial characters in a command. The commands that start with the entered letters are then displayed. For example, if you enter **dis?**, the device displays all commands that start with **dis**.

If you enter a space after entering partial characters in a command and then enter a question mark (?), the device attempts to identify the command corresponding to the entered information and displays other parameters of the command. For example, after you enter **dis ?**, if only one command starts with **dis**, the device displays the parameters of the **display** command. However, if there are other commands that start with **dis**, the device displays an error message.

You can also use the **Tab** key on the keyboard to complete the command. For example, after you enter **dis**, press **Tab** to expand it to **display**. If there are multiple commands that start with **dis**, the device displays the commands cyclically.

You can use the abbreviation of a command if no ambiguity occurs. For example, **display** can be abbreviated to **dis** or **disp**, and **interface** can be abbreviated to **int** or **inter**.

### 3.5 Changing Device Names

When configuring devices, you are advised to assign distinct names to facilitate differentiation. Change device names based on the lab topology.

Change the name of the router to R1. To make the change, run the **system-view** command to enter the system view first.

```
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sy
```



```
[Huawei]sysname R1
[R1]
```

### 3.6 Configuring Login Information

Configure login messages to display a warning or prompt when users log in. Run the **header shell information** command to configure login information.

```
[R1]header shell information "Welcome to the Huawei certification lab."
```

Exit the CLI and log in to the CLI again. Check whether the login information is modified.

```
[R1]quit
<R1>
<R1>quit

Configuration console exit, please press any key to log on

Welcome to the Huawei certification lab.
<R1>
<R1>
```

### 3.7 Setting Console Port Parameters

The console port allows access without a password by default. Users can directly connect and configure the device.

To avoid risks, you can configure password authentication for console port-based login and set a ciphertext password such as **Huawei@123**.

An idle timeout interval refers to the automatic exit from the configuration interface after a certain period of inactivity. Upon logging in again, the system prompts you to enter the password for verification.

Set the idle timeout interval to 20 minutes. The default value is 10 minutes.

```
[R1]user-interface console 0
[R1-ui-console0]authentication-mode password
[R1-ui-console0]set authentication password cipher
Warning: The "password" authentication mode is not secure, and it is strongly recommended to use
"aaa" authentication mode.
Enter Password(<8-128>):
Confirm password:
[R1-ui-console0]idle-timeout 20 0
[R1-ui-console0]
```

Run the **display this** command to check the configuration result.

```
[R1-ui-console0]display this
[V200R007C00SPC600]
#
user-interface con 0
 authentication-mode password
 set authentication password
 cipher %^%#[cR8Y%Qf_6Ra=OPEu'SFa*b$4hjW[O!/dX,6>9xW:ZQMPh6R1SbJt2SW`Y]:%^%#
 idle-timeout 20 0
```



```
user-interface vty 0
 authentication-mode aaa
 user privilege level 15
user-interface vty 1 4
#
return
```

Exit the system, and then log in to the system using the new password. Note that you also need to configure a password when starting the router for the first time after initialization.

```
[R1-ui-console0]return
<R1>quit
```

Configuration console exit, please press any key to log on

```
Login authentication
Password:
Welcome to Huawei certification lab
<R1>
```

### 3.8 Configuring Interface IP Addresses and Descriptions

Configure an IP address for GigabitEthernet 0/0/0 on R1. Set a value in dotted decimal notation (for example, 255.255.255.0). You can also configure a subnet mask based on the subnet mask prefix length.

```
[R1]int g0/0/0
[R1-GigabitEthernet0/0/0]ip address 172.16.10.1 24
[R1-GigabitEthernet0/0/0]description This interface connects to PC
[R1-GigabitEthernet0/0/0]
```

Run the **display this** command in the interface view to check the configuration.

```
[R1-GigabitEthernet0/0/0]display this
[V200R003C00]
#
interface GigabitEthernet0/0/0
 description This interface connects to PC
 ip address 172.16.10.1 255.255.255.0
#
return
[R1-GigabitEthernet0/0/0]quit
[R1]
```

Run the **display interface** command to check interface information.

```
[R1]display interface GigabitEthernet0/0/0
GigabitEthernet0/0/0 current state : UP
Line protocol current state : UP
Last line protocol up time : 2025-04-03 09:08:04 UTC+08:00
Description:This interface connects to PC
Route Port,The Maximum Transmit Unit is 1500
Internet Address is 172.16.10.1/24
IP Sending Frames' Format is PKTFMT_ETHNT_2, Hardware address is 00e0-fcfe-213f
Last physical up time : 2025-04-03 09:08:04 UTC+08:00
```



```
Last physical down time : 2025-04-03 08:43:26 UTC-08:00
Current system time: 2025-04-03 09:09:05+08:00
Port Mode: FORCE COPPER
Speed : 1000, Loopback: NONE
Duplex: FULL, Negotiation: ENABLE
Mdi : AUTO
Last 300 seconds input rate 8 bits/sec, 0 packets/sec
Last 300 seconds output rate 8 bits/sec, 0 packets/sec
Input peak rate 688 bits/sec,Record time: 2025-04-03 09:09:02
Output peak rate 688 bits/sec,Record time: 2025-04-03 09:09:02
```

```
Input: 6 packets, 430 bytes
```

|            |    |              |   |
|------------|----|--------------|---|
| Unicast:   | 6, | Multicast:   | 0 |
| Broadcast: | 0, | Jumbo:       | 0 |
| Discard:   | 0, | Total Error: | 0 |
| CRC:       | 0, | Giants:      | 0 |
| Jabbers:   | 0, | Throttles:   | 0 |
| Runts:     | 0, | Symbols:     | 0 |
| Ignoreds:  | 0, | Frames:      | 0 |

```
Output: 7 packets, 490 bytes
```

|                  |    |                      |   |
|------------------|----|----------------------|---|
| Unicast:         | 5, | Multicast:           | 0 |
| Broadcast:       | 2, | Jumbo:               | 0 |
| Discard:         | 0, | Total Error:         | 0 |
| Collisions:      | 0, | ExcessiveCollisions: | 0 |
| Late Collisions: | 0, | Deferreds:           | 0 |

```
Input bandwidth utilization threshold : 100.00%
Output bandwidth utilization threshold: 100.00%
Input bandwidth utilization : 0%
Output bandwidth utilization : 0%
```

The command output on R1 shows that the physical state and protocol state of the interface are both Up, indicating that the physical layer and data link layer function properly.

Set the IP address of the PC to 172.16.10.2/24. After the configuration is complete, run the **ping** command to test the connectivity between R1 and the PC.

```
[R1]ping 172.16.10.2
PING 172.16.10.2: 56 data bytes, press CTRL_C to break
Reply from 172.16.10.2: bytes=56 Sequence=1 ttl=128 time=20 ms
Reply from 172.16.10.2: bytes=56 Sequence=2 ttl=128 time=30 ms
Reply from 172.16.10.2: bytes=56 Sequence=3 ttl=128 time=10 ms
Reply from 172.16.10.2: bytes=56 Sequence=4 ttl=128 time=30 ms
Reply from 172.16.10.2: bytes=56 Sequence=5 ttl=128 time=10 ms

--- 172.16.10.2 ping statistics ---
5 packet(s) transmitted
5 packet(s) received
0.00% packet loss
```



```
round-trip min/avg/max = 10/20/30 ms
```

```
[R1]
```

### 3.9 Checking the File List on a Device

Run the **dir** command in the user view to check the file list in the current directory.

```
<R1>dir
```

```
Directory of flash:/
```

| Idx | Attr | Size(Byte) | Date        | Time(LMT) | FileName         |
|-----|------|------------|-------------|-----------|------------------|
| 0   | drw- | -          | Apr 02 2025 | 08:26:23  | dhcp             |
| 1   | -rw- | 121,802    | May 26 2014 | 09:20:58  | portalpage.zip   |
| 2   | -rw- | 2,263      | Apr 02 2025 | 08:26:18  | statemach.efs    |
| 3   | -rw- | 828,482    | May 26 2014 | 09:20:58  | sslvpn.zip       |
| 4   | -rw- | 249        | Apr 02 2025 | 11:08:14  | private-data.txt |
| 5   | -rw- | 591        | Apr 02 2025 | 11:31:39  | vrpcfg.zip       |

```
1,090,732 KB total (784,456 KB free)
```

```
<R1>
```

### 3.10 Managing the Configuration File of a Device

Run the **display saved-configuration** command to check the saved configuration file.

```
<R1>display saved-configuration
```

```
There is no correct configuration file in FLASH
```

No configuration file is saved in the system. Run the **save** command to save the current configuration file.

```
<R1>save
```

```
The current configuration will be written to the device.
```

```
Are you sure to continue? (y/n)[n]:y
```

```
It will take several minutes to save configuration file, please wait.....
```

```
Configuration file had been saved successfully
```

```
Note: The configuration file will take effect after being activated
```

Run the **display saved-configuration** command again to check saved configuration information.

```
<R1>display saved-configuration
```

```
[V200R003C00]
```

```
#
```

```
sysname R1
```

```
header shell information "Welcome to the Huawei certification lab."
```

```
#
```

```
snmp-agent local-engineid 800007DB0300000000000000
```

```
snmp-agent
```

```
#
```

```
clock timezone Local add 08:00:00
```

```
#
```

```
portal local-server load portalpage.zip
```

```
#
```



```
drop illegal-mac alarm
#
set cpu-usage threshold 80 restore 75
#
aaa
 authentication-scheme default
 authorization-scheme default
 accounting-scheme default
 domain default
 domain default_admin
 local-user admin password cipher %$%$K8m.Nt84DZ}e#<0`8bmE3Uw}%$%$
 local-user admin service-type http
#
firewall zone Local
 priority 15
#
interface GigabitEthernet0/0/0
 description This interface connects to PC
 ip address 172.16.10.1 255.255.255.0
#
interface GigabitEthernet0/0/1
#
interface GigabitEthernet0/0/2
#
interface NULL0
#
user-interface con 0
 authentication-mode password
user-interface vty 0 4
user-interface vty 16 20
#
wlan ac
#
return
<R1>
```

Run the **display current-configuration** command to check the current configuration.

```
<R1>display current-configuration
[V200R003C00]
#
 sysname R1
 header shell information "Welcome to the Huawei certification lab."
#
 snmp-agent local-engineid 800007DB0300000000000000
 snmp-agent
#
 clock timezone Local add 08:00:00
#
 portal local-server load portalpage.zip
#
 drop illegal-mac alarm
```



```
#
 set cpu-usage threshold 80 restore 75
#
aaa
 authentication-scheme default
 authorization-scheme default
 accounting-scheme default
 domain default
 domain default_admin
 local-user admin password cipher %$%$K8m.Nt84DZ}e#<0`8bmE3Uw}%$%$
 local-user admin service-type http
#
 firewall zone Local
  priority 15
#
 interface GigabitEthernet0/0/0
  description This interface connects to PC
  ip address 172.16.10.1 255.255.255.0
#
 interface GigabitEthernet0/0/1
#
 interface GigabitEthernet0/0/2
#
 interface NULL0
#
 user-interface con 0
  authentication-mode password
 user-interface vty 0 4
 user-interface vty 16 20
#
 wlan ac
#
 return
<R1>
```

A router can have multiple configuration files stored. Run the **display startup** command to check the configuration file used for next startup.

```
<R1>display startup
MainBoard:
  Startup system software:          null
  Next startup system software:     null
  Backup system software for next startup: null
  Startup saved-configuration file:  flash:/vrpcfg.zip
  Next startup saved-configuration file: flash:/vrpcfg.zip
  Startup license file:             null
  Next startup license file:        null
  Startup patch package:            null
  Next startup patch package:       null
  Startup voice-files:              null
  Next startup voice-files:         null
<R1>
```



Delete the configuration file in the flash memory.

```
<R1>reset saved-configuration
This will delete the configuration in the flash memory.
The device configurations will be erased to reconfigure.
Are you sure? (y/n)[n]:y
Clear the configuration in the device successfully.
```

### 3.11 Restarting a Device

Run the **reboot** command to restart a router.

```
<R1>reboot
Info: The system is now comparing the configuration, please wait.
Warning: All the configuration will be saved to the next startup configuration. Continue ? [y/n]:n
System will reboot! Continue ? [y/n]:y
Info: system is rebooting ,please wait...
```

The system prompts you to save the current configurations. Make your decision based on experiment needs. If you are not sure whether to save the current configurations, do not save them.

### 3.12 Remotely Managing a Device Using Telnet

Telnet provides an interactive interface, allowing a terminal user to remotely log in to any device that can function as a Telnet server. Telnet allows users to control a device remotely just like through its local console port. The remote Telnet server and terminal do not need to be directly connected; However, it is necessary to ensure that they can communicate with each other. Telnet enables users to remotely manage and maintain devices.

Telnet runs in client/server mode. Telnet relies on TCP and operates on port 23 by default. This port allows the server to connect to client devices.

When configuring Telnet login, you must configure the authentication mode. Otherwise, you will fail to log in to the device.

Telnet supports two authentication modes: AAA and password.

- If AAA authentication is configured, you need to enter the correct user name and password to log in to a device.
- If password authentication is configured, you need to enter the correct password to log in to a device.

A network device acting as a Telnet server typically employs a password authentication mechanism to verify users connecting to the VTY interface.

Virtual Type Terminal (VTY) is an interface used by network devices to manage and monitor users logging in through Telnet. A network device allocates a VTY interface to each Telnet user. By default, an ARG3 series router supports a maximum of five Telnet users. **vtty 0 4** indicate VTY0, VTY1, VTY2, VTY3, and VTY4. To increase the number of Telnet users, you can run the **user-interface maximum-vty** command to adjust the number of VTY interfaces.

You can run the **authentication-mode password** command to configure the VTY interface to authenticate users using passwords.

Note: The **set authentication password cipher** command varies according to the VRP version. For example, after you enter the command, some versions require you to press **Enter**



before entering a password, while others allow you to enter a password directly. For details, see the related VRP documentation.

Configure the Telnet server function on R1.

```
<R1>system-view
Enter system view, return user view with Ctrl+Z.
[R1]user-interface vty 0 4
[R1-ui-vty0-4]
[R1-ui-vty0-4]authentication-mode password
Please configure the login password (maximum length 16):Huawei@123
[R1-ui-vty0-4]user privilege level 3
[R1-ui-vty0-4]quit
[R1]
```

Telnet R1 from the PC, open the CMD window on the PC, and run the following commands:

```
C:\Users\Administrator> telnet 172.16.10.1
Press CTRL_] to quit telnet mode
Trying 172.16.10.1 ...
Connected to 172.16.10.1 ...

Login authentication

Password:Huawei@123
Welcome to the Huawei certification lab.
<R1>system-view
Enter system view, return user view with Ctrl+Z.
[R1]display version
Huawei Versatile Routing Platform Software
VRP (R) software, Version 5.130 (AR2200 V200R003C00)
Copyright (C) 2011-2012 HUAWEI TECH CO., LTD
Huawei AR2220 Router uptime is 0 week, 0 day, 0 hour, 58 minutes
BKP 0 version information:
1. PCB      Version   : AR01BAK2A VER.NC
2. If Supporting PoE : No
3. Board    Type      : AR2220
4. MPU Slot Quantity : 1
5. LPU Slot Quantity : 6

MPU 0(Master) : uptime is 0 week, 0 day, 0 hour, 58 minutes
MPU version information :
1. PCB      Version   : AR01SRU2A VER.A
2. MAB      Version   : 0
3. Board    Type      : AR2220
4. BootROM  Version   : 0

[R1]quit
<R1>quit

Configuration console exit, please retry to log on

The connection was closed by the remote host
```



```
C:\Users\Administrator>
```

### 3.13 Remotely Managing a Device Using SSH

Telnet lacks a secure authentication mode and uses TCP to transmit data in cleartext, which brings great security risks. Compared with Telnet, SSH guarantees secure file transfer on a traditional insecure network by authenticating clients and encrypting data in bidirectional mode.

The SSH user authentication protocol provides the following two authentication methods:

- Password authentication: The client uses the username and password to log in to the server for user authentication.
- Public key authentication: The server uses the public key to decrypt the digital signature of the client for user authentication.

#### Step 1 Clear the configurations of R1 and restart R1.

```
<R1>reset saved-configuration
This will delete the configuration in the flash memory.

The device configuration
ns will be erased to reconfigure.

Are you sure? (y/n)[n]:y
Clear the configuration in the device successfully.

<R1>reboot
Info: The system is comparing the configuration, please wait.
Warning: All the configuration will be saved to the next startup configuration.
Continue ? [y/n]:n
System will reboot! Continue ? [y/n]:y
Info: system is rebooting ,please wait...
```

#### Step 2 Reconfigure R1.

```
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname R1
[R1]int g0/0/0
[R1-GigabitEthernet0/0/0]ip address 172.16.10.1 24
[R1-GigabitEthernet0/0/0]quit
[R1]ping 172.16.10.2
  PING 172.16.10.2: 56 data bytes, press CTRL_C to break
    Reply from 172.16.10.2: bytes=56 Sequence=1 ttl=128 time=80 ms
    Reply from 172.16.10.2: bytes=56 Sequence=2 ttl=128 time=20 ms
    Reply from 172.16.10.2: bytes=56 Sequence=3 ttl=128 time=20 ms
    Reply from 172.16.10.2: bytes=56 Sequence=4 ttl=128 time=20 ms
    Reply from 172.16.10.2: bytes=56 Sequence=5 ttl=128 time=30 ms

--- 172.16.10.2 ping statistics ---
  5 packet(s) transmitted
  5 packet(s) received
```



```
0.00% packet loss
round-trip min/avg/max = 20/34/80 ms
```

```
[R1]
```

### Step 3 Configure R1 as an SSH server.

```
<R1>system-view
Enter system view, return user view with Ctrl+Z.
[R1]aaa
[R1-aaa]local-user user01 password cipher Huawei12#$
Info: Add a new user.
[R1-aaa]local-user user01 service-type ssh
[R1-aaa]quit
[R1]
[R1]stelnet server enable
Info: Succeeded in starting the STELNET server.
[R1]undo ssh server compatible-ssh1x enable
Disable SSHv1.
[R1]ssh user user01 authentication-type password
Authentication type setted, and will be in effect next time
[R1]
[R1]user-interface vty 0 4
[R1-ui-vty0-4]authentication-mode aaa
[R1-ui-vty0-4]user privilege level 3
[R1-ui-vty0-4]protocol inbound ssh
[R1-ui-vty0-4]quit
[R1]
```

### Step 4 Log in to R1 using SSH.

```
[R1]stelnet 172.16.10.1
Please input the username:user01
Trying 172.16.10.1 ...
Press CTRL+K to abort
Connected to 172.16.10.1 ...
Error: Failed to verify the server's public key.
Please run the command "ssh client first-time enable"to enable the first-time access function and try again.
[R1]
[R1]
[R1]ssh client first-time enable
[R1]
[R1]stelnet 172.16.10.1
Please input the username:user01
Trying 172.16.10.1 ...
Press CTRL+K to abort
Connected to 172.16.10.1 ...
The server is not authenticated. Continue to access it? (y/n)[n]:y
Apr  3 2025 10:17:01-08:00 R1 %01SSH/4/CONTINUE_KEYEXCHANGE(I)[1]:The server has not been authenticated in the process of exchanging keys. When deciding whether to continue, the user chose Y.
[R1]
```



```
Save the server's public key? (y/n)[n]:y
The server's public key will be saved with the name 172.16.10.1. Please wait...

Apr  3 2025 10:17:05-08:00 R1  %%01SSH/4/SAVE_PUBLICKEY(I)[2]:When deciding whether to save the server's public key 172.16.10.1, the user chose Y.
[R1]
Enter password:Huawei12#$
<R1>
<R1>system-view
Enter system view, return user view with Ctrl+Z.
[R1]display ssh server session

-----
Conn    Ver    Encry    State  Auth-type    Username
-----
VTY 0   2.0    AES      run    password     user01
-----

[R1]

----End
```

## 4 Verification

1. Check whether STelnet is enabled on R1.

```
[R1]display ssh server status
SSH version                :1.99
SSH connection timeout     :60 seconds
SSH server key generating interval :0 hours
SSH Authentication retries  :3 times
SFTP Server                 :Disable
Stelnet server              :Enable
[R1]
```

2. Check the current SSH login user.

```
[R1]display ssh server session

-----
Conn    Ver    Encry    State  Auth-type    Username
-----
VTY 0   2.0    AES      run    password     user01
-----

[R1]
```

## 5 Quiz

1. What is the maximum number of users allowed to log in to a Huawei device through the console port concurrently?  
A Huawei device allows only one user to log in through the console port at a time. Therefore, the console user ID is fixed at 0.
2. How do I specify the configuration file for next startup if a device has multiple configuration files?



You can save the configuration file with a specified name instead of the default name **vrp.cfg** on the router or switch. To specify a configuration file for next startup, run the **startup saved-configuration** [ *configuration-file-name* ] command. The configuration file name should contain both the file name and extension.

3. If the Telnet service has been configured on a network device but a user still cannot realize remote access, how can I determine the cause?

If the Telnet connection cannot be established, check whether the device is reachable. If the device is reachable, check whether the entered password is correct. If the password is correct, check whether the number of Telnet users has reached the upper limit. To increase the number of users, run the **user-interface maximum-vty <0-15>** command. **0-15** indicates the number of supported users.

4. After the SSH configuration is complete, use PuTTY to log in to the router in SSH mode.